

PEE question set 1

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1 Theoretical questions

1.1 Question 1

Given a HMM, and a training sample, this exercise required us to perform a single iteration over the estimation algorithm. To do so, we have generalized the expression referenced in the theory document to HMM. Thus, the learning algorithm will result in the following:

$$\theta_{ij} = \frac{\sum_{x \in D} \frac{1}{P_\theta(x)} \sum_{\forall dx} N(A \rightarrow BC, dx) P_\theta(x, dx)}{\sum_{x \in D} \frac{1}{P_\theta(x)} \sum_{\forall dx} N(A, dx) P_\theta(x, dx)}$$

To perform this calculation we have to get first the probability for both training samples:

$$P_\theta(ba) = q(1|\#) \times e(c|1) \times q(3|1) \times e(a|3) = 1 \times 0.3 \times 0.3 \times 1 \times 0.6 = 0.054$$

For the second training sample we find that there are three different derivation trees that can produce it, so the complete probability for it will be the sum of the probabilities given by each tree.

$$P_\theta(ccca) = P_\theta(ccca|t1) + P_\theta(ccca|t2) + P_\theta(ccca|t3) = 0.000441 + 0.002646 + 0.00247 = 0.005557$$

The exercise asks only for the calculations for the transitions between state 1 and 2. That corresponds to $P_\theta(1|2)$ and $P_\theta(2|1)$. Thus, for one iteration the calculus will be:

$$P_\theta(2|1) = \frac{\frac{1}{0.054} \times 0 \times 0.054 + \frac{1}{0.005557} \times (1 \times 0.000441 + 1 \times 0.002646 + 0 \times 0.00247)}{\frac{1}{0.054} \times 1 \times 0.054 + \frac{1}{0.005557} \times (2 \times 0.000441 + 2 \times 0.002646 + 3 \times 0.00247)} = 0.16128$$

$$P_\theta(1|2) = \frac{\frac{1}{0.054} \times 0 \times 0.054 + \frac{1}{0.005557} \times (1 \times 0.000441 + 0 \times 0.002646 + 0 \times 0.00247)}{\frac{1}{0.054} \times 0 \times 0.054 + \frac{1}{0.005557} \times (1 \times 0.000441 + 1 \times 0.002646 + 0 \times 0.00247)} = 0.14285$$

The complete results for the first algorithm iteration are shown below:

$P_\theta(3|1) = 0.4424$
 $P_\theta(2|1) = 0.16128$
 $P_\theta(1|1) = 0.39632$
 $P_\theta(3|2) = 0.85714$
 $P_\theta(1|2) = 0.14285$
 $P_\theta(3|3) = 0.0$
 $P_\theta(\#|3) = 1.0$
 $P_\theta(a|1) = 0.0$
 $P_\theta(b|1) = 0.29032$
 $P_\theta(c|1) = 0.70968$
 $P_\theta(a|2) = 0.0$
 $P_\theta(b|2) = 0.0$
 $P_\theta(c|2) = 1.0$
 $P_\theta(a|3) = 1.0$
 $P_\theta(b|3) = 0.0$
 $P_\theta(c|3) = 0.0$

The resulting HMM can be seen in the figure (1a) corresponding to the fifth question of this set. Which was asking precisely that. In that figure the emission weights can not be seen, but they're displayed in the previous equations.

1.2 Question 2

As we can see in the results of the previous question, some emission parameters become 0. More precisely, become 0 the emission of symbols 'a' and 'b' in the second state ($P_\theta(a|2)$ and $P_\theta(b|2)$) and the transition from state 3 to himself. To avoid this issue, we simply have to provide a training sample capable of emitting either 'a' or 'b' at the this state and with more than one 'a' at the end. This sequence: "cbaa" solves the problem. To prove this is a solution, we can train the HMM from scratch with this new training sample and see how the emission probabilities differ from 0.

$D = ccca, ba, cbaa$

We see that this new sequence has three parsing trees:

$t1 : \# \rightarrow 1; 1 \rightarrow 1; 1 \rightarrow 2; 2 \rightarrow 3; 3 \rightarrow \#$

$t2 : \# \rightarrow 1; 1 \rightarrow 2; 2 \rightarrow 3; 3 \rightarrow 3; 3 \rightarrow \#$

$t3 : \# \rightarrow 1; 1 \rightarrow 1; 1 \rightarrow 3; 3 \rightarrow 3; 3 \rightarrow \#$

$$P_\theta(cbaa) = P_\theta(cbaa|t1) + P_\theta(cbaa|t2) + P_\theta(cbaa|t3) = 0.04536 + 0.003402 + 0.003024 = 0.051786$$

The new HMM weights will be:

$P_\theta(3|1) = 0.2942$

$P_\theta(2|1) = 0.2783$

$P_\theta(1|1) = 0.4275$

$P_\theta(3|2) = 0.947$

$P_\theta(1|2) = 0.053$

$P_\theta(3|3) = 0.0397$

$P_\theta(\#|3) = 0.9603$

$P_\theta(a|1) = 0.0$

$P_\theta(b|1) = 0.3596$

$P_\theta(c|1) = 0.6404$

$P_\theta(a|2) = 0.5851$

$P_\theta(b|2) = 0.0439$

$P_\theta(c|2) = 0.3710$

$P_\theta(a|3) = 1.0$

$P_\theta(b|3) = 0.0$

$P_\theta(c|3) = 0.0$

As we can see, no emission in state 2 has now zero probability, so the string we have suggested will solve the problem faced in Question 1.

2 Practical questions

2.1 Question 3

We have parsed the *TextHMM* file with HMML and the output has been the following:

A.V. Aho and J.D. Ullman The theory of parsing translation and compiling. Volumen I: parsing 1972 Prentice-Hall

P.F. Brown The acoustic-Modeling Problem in Automatic Speech Recognition 1987 Carnegie-Mellon University Ph.D. dissertation

L.E. Baum An inequality and Associated Maximization Technique in Statistical Estimation for Probabilistic Functions of Markov Processes Inequalities 1972 3 1-13 8

J. Kupiec Hidden Markov Estimation for Unrestricted Stochastic Context-Free Grammars Proc. of ICASSP'92 Vol. 1 1992 177-180

To see it clearly, we have developed a parser script that transforms this raw output in bib format:

```
@inproceedings{
  author = { A.V. Aho and J.D. Ullman },
  title = { The theory of parsing translation and compiling. Volumen I: parsing },
  year = { 1972 },
  publisher = { Prentice-Hall }
}
@inproceedings{
  author = { P.F. Brown },
  title = { The acoustic-Modeling Problem in Automatic },
  journal = { Speech Recognition },
  school = { Carnegie-Mellon University },
  type = { Ph. D. dissertation },
  year = { 1987 }
}
@inproceedings{
  author = { L.E. Baum },
  title = { An inequality and Associated Maximization Technique in Statistical Estimation for Probabilistic Functions of Markov Processes },
  journal = { Inequalities },
  volume = { 3 },
  number = { 1 }
```

```

        year =          { 1972  }
        pages =         { 8    }
    }
    @inproceedings{
        author =         { J. Kupiec  }
        title =          { Hidden Markov Estimation for Unrestricted Stochastic Context-Free Grammars  }
        year = { 1992    }
        booktitle =       { Proc. of ICASSP'92 Vol.  }
        pages =          { 1 177 180  }
    }
}

```

2.2 Question 4

As done in the previous question, we're going to show the raw output obtained by calling the script with the model trained 10 iterations:

```

A.V.@0 Aho@0 and@0 J.D.@0 Ullman@0 The@1 theory@1 of@1 parsing@1 translation@1 and@1 compiling.@1 Volumen@1 I:@1 parsing@1
1972@1 Prentice-Hall@3

P.F.@0 Brown@0 The@1 acoustic-Modeling@1 Problem@1 in@1 Automatic@1 Speech@11 Recognition@11 1987@2 Carnegie-Mellon@16
University@16 Ph.@17 D.@17 dissertation@17

L.E.@0 Baum@0 An@1 inequality@1 and@1 Associated@1 Maximization@1 Technique@1 in@1 Statistical@1 Estimation@1 for@1
Probabilistic@1 Functions@1 of@1 Markov@1 Processes@1 Inequalities@11 1972@2 3@12 1@13 8@8

J.@0 Kupiec@1 Hidden@1 Markov@1 Estimation@1 for@1 Unrestricted@1 Stochastic@1 Context-Free@1 Grammars@1 Proc.@4 of@4
ICASSP'92@2 Vol.@12 1@13 1992@2 177@8 180@8

```

And its corresponding bib format to see it more clearly.

```

@inproceedings{
    author =          { A.V. Aho and J.D. Ullman  }
    title =           { The theory of parsing translation and compiling. Volumen I: parsing 1972  }
    publisher =       { Prentice-Hall  }
}
@inproceedings{
    author =          { P.F. Brown  }
    title =           { The acoustic-Modeling Problem in Automatic  }
    journal =         { Speech Recognition  }
    school =          { Carnegie-Mellon University  }
    type =            { Ph. D. dissertation  }
    year =            { 1987  }
}
@inproceedings{
    author =          { L.E. Baum  }
    title =           { An inequality and Associated Maximization Technique in Statistical Estimation for
    Probabilistic Functions of Markov Processes  }
    journal =         { Inequalities  }
    volume =          { 3  }
    number =          { 1  }
    year =            { 1972  }
    pages =           { 8  }
}
@inproceedings{
    author =          { J.  }
    title =           { Kupiec Hidden Markov Estimation for Unrestricted Stochastic Context-Free Grammars  }
    volume =          { Vol.  }
    number =          { 1  }
    year =            { ICASSP'92 1992  }
    booktitle =       { Proc. of  }
    pages =           { 177 180  }
}

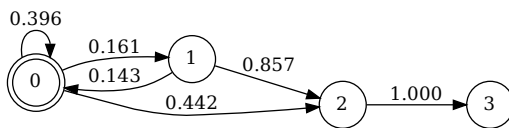
```

As we can see comparing both outputs, “P.F. Brown” and “L.E. Baum” have remained the same. But in “Aho & Ullman” and “J. Kupiec” we can see some differences. At the first reference we see how the new model has failed to classify the year correctly. Miss-interpreting it as if it was part of the title. At the last one we can see another miss-interpretations, classifying the author’s name as part of the title. We can see also some classification failures regarding to the year.

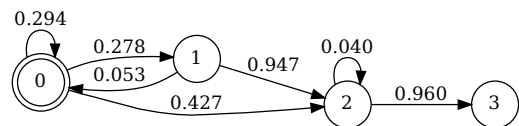
As seen, training the HMM with this 10 iterations has produced a fairly worse model. This could have been caused by the nature of the training samples, having they mostly references with more than one author. Also in most of the cases, the year comes before another set of numbers. That differences between the training and test samples have caused that the model could not produce the correct output after 10 iterations of training.

2.3 Question 5

After the training process the HMM obtained have the following form:



(a) Model Question 1



(b) Model Question 1